

Element I

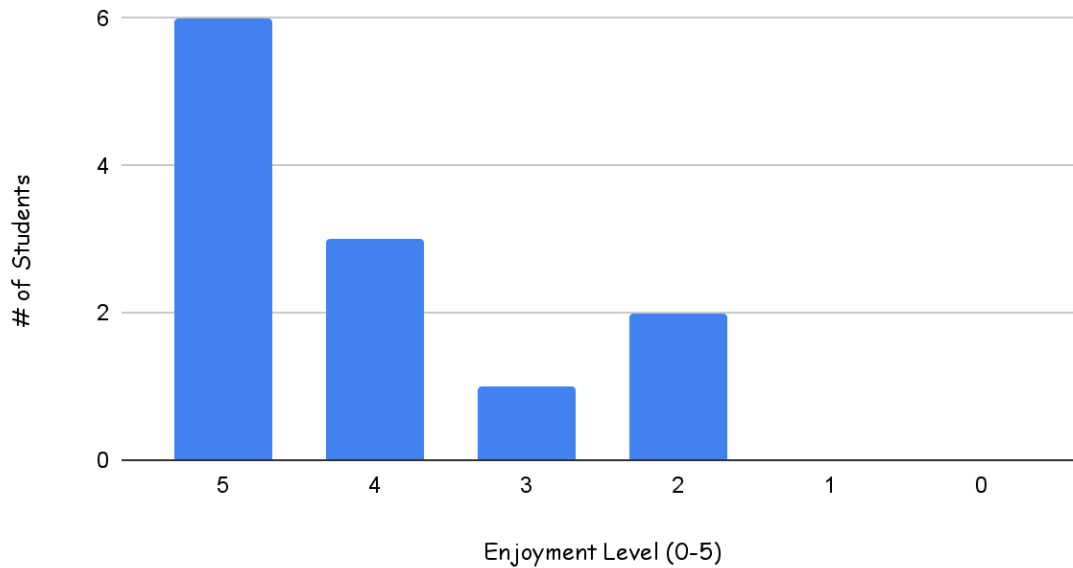
The importance of testing all of these design requirements lies in the satisfaction of our customer and stakeholders. Once again, our high priorities are that the product is safe, aesthetic, durable, entertaining, and adaptable. All of the ways we will test this will be in a classroom environment in order to most effectively mimic the setting which the lights will be incorporated into. Additionally, other testing procedures will anticipate conditions that the LED lights will be exposed to, as seen in the easy to use testing procedures.

These tests are crucial to the design process because we need a specific and measurable way to determine if our team designed an effective product. This is also an important part of the process so that we know exactly where our product falls short and where it has the potential to grow. This way, our assessment is unbiased and simply based on how well our product met our requirements.

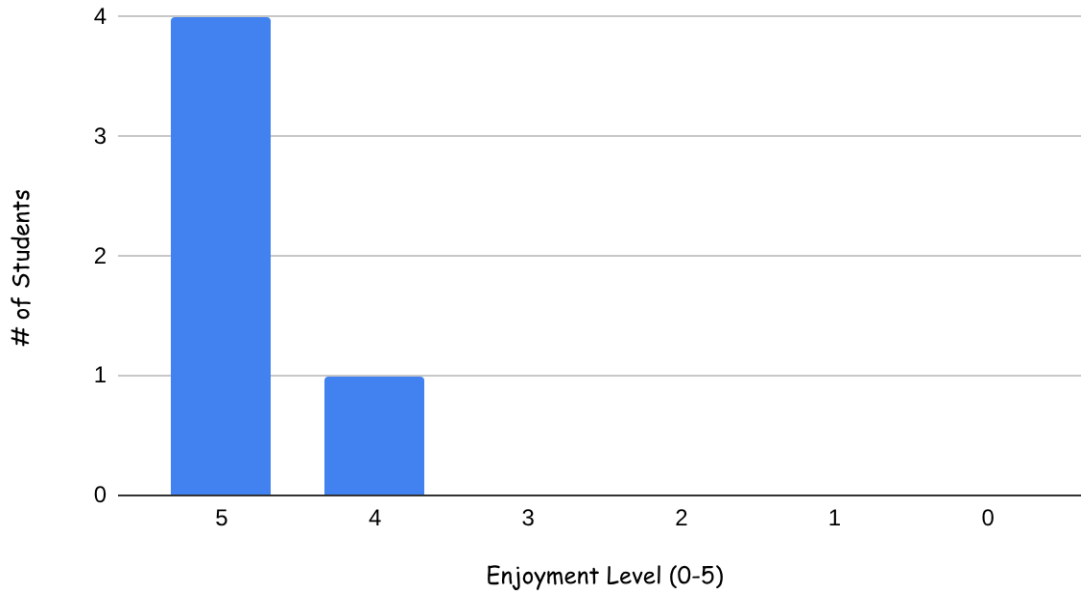
We made sure to talk to teachers and students (stakeholders) in order to understand their impression of our product as of right now:

Notes, data, graphs, and tables from our tests:

Student's Feedback



Teacher's Feedback



These charts show the rating the students and teachers gave the lights after we installed them. We did a simple sample, with the first graph 1 having an average of 4.08 and graph 2 having an average of 4.8. These results are satisfactory, but there could be some bias because they may have feared about hurting our feelings.



Priority requirement	Gist of testing procedure	Results
Safety	We exposed the kids to the lights at the height where we believed they would be out of reach	Lights were out of reach of children, (above 5 feet) except for wires coming out of outlets, which could be pulled on.
Aesthetic/Creates a diverse mood	Place lights out in a miniature model of the classroom, and try out different color schemes and patterns using app	We were able to zone the lights and create a variety of moods using the zones and a united pattern
Durability	Overtime we will monitor the condition of the lights and note any damaged lights, lack in efficiency, and the lights falling off.	N/A
Entertaining and Effectiveness	We ran some features of the lights in order to see the reactions of the kids and adults.	The students were engaged with the lights' features and teachers were able to separate the class into zones. We asked them to rate the light on a scale of 0-5 and got an average of 4.08, which passed our test.
Adaptability to different	We spoke to our primary	After using the lights, our

needs	stakeholder to ensure our lights met the needs of the classroom aids.	primary stakeholders reported that the lights helped aid all the needs that the classroom had.
Ease of use	We gave our primary stakeholders five different tasks on the light to see if they would be able to complete it without assistance	The primary stakeholders were able to complete 3/5 tasks without assistance, and could do the other two with some help

Detail of design analysis: We went into the classroom and took measurements of the dimensions of the room so we knew how many strips of lights we needed and how long. The room was approximately 30' by 17'. Therefore, we bought 3 bluetooth RGB LED lights from Govee (100ft, 130ft, and 32.8ft) to split up between different sections of the room.

Summary of data implications: Overall, the prototype performed well against all of the testing plans. However, there were some aspects of the design that we could improve on. Most of the tests that didn't fully pass had easy ways to improve the design. Next steps to improve the design would include finding a way to cover the lugs at the outlet and creating a system to help the teachers use the lights more easily.